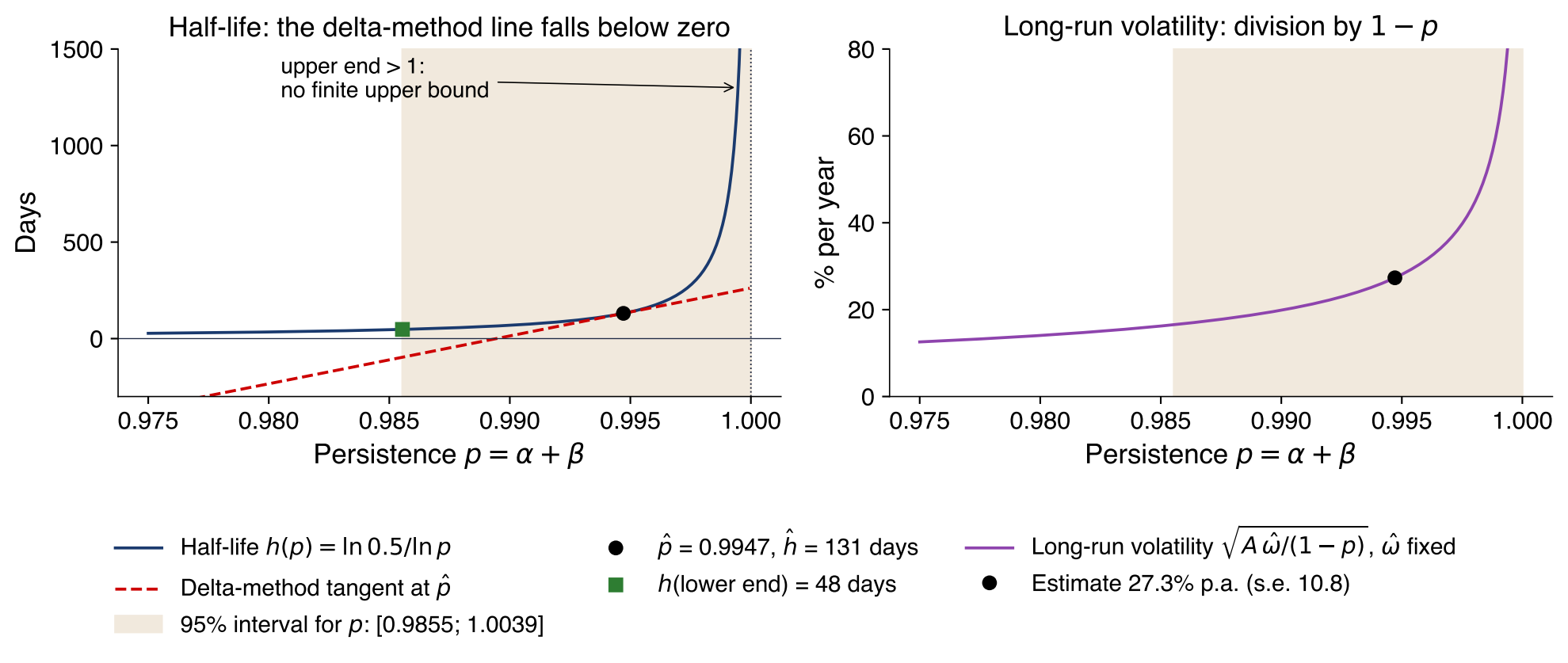
Half-life: the delta-method line falls below zero
upper end > 1:
no finite upper bound
Days
1500
1000
500
0
0.975
0.980
0.985
0.990
0.995
1.000
Persistence $p = \alpha + \beta$
Long-run volatility: division by $1 - p$
% per year
80
60
40
20
0
0.975
0.980
0.985
0.990
0.995
1.000
Persistence $p = \alpha + \beta$
Half-life $h(p) = \ln 0.5 / \ln p$
Delta-method tangent at $\hat{p}$
95% interval for p: [0.9855; 1.0039]
$\hat{p} = 0.9947$, $\hat{h} = 131$ days
h(lower end) = 48 days
Long-run volatility $\sqrt{A\,\hat{\omega}/(1-p)}$, $\hat{\omega}$ fixed
Estimate 27.3% p.a. (s.e. 10.8)